

PRODUCTION-READY MASTER SPECIFICATION

VETRALINK PRO

Integrated Livestock Intelligence, Farm ERP & Tele-Veterinary Ecosystem

Author: Lead Veterinary Consultant & Chief Architect

Target Market: Dairy, Cattle, Goat & Poultry Farmers

Release Baseline: Version 1.0 (Production Blueprint)

1. Executive Summary & Value Proposition

VETRALINK PRO is a unified, cloud-native AgTech platform engineered to bridge the critical gap between veterinary science, daily farm operations, and commercial farm profitability. While agriculture and livestock rearing represent the backbone of rural and semi-urban economies worldwide, the majority of small-to-medium enterprise (SME) farm owners operate in silos—relying on fragmented paper logs, unverified social media advice, and reactive, emergency-only veterinary interventions.

By synthesizing **Educational Content** (LMS), **Validated Digital Assets** (eBooks & Excel Toolkits), **Multi-Species Farm SaaS ERP**, and **Direct Tele-Veterinary Consultations** into a cohesive ecosystem, **VETRALINK PRO** transforms traditional livestock farming into a data-driven, preventive, and highly profitable enterprise.

35-40%

REDUCTION IN
PREVENTABLE
LIVESTOCK MORTALITY

18-22%

FEED OPTIMIZATION &
COST REDUCTION

100%

DIGITAL TRACEABILITY &
HEALTH HISTORY

3.5x

FASTER VET DIAGNOSTIC
RESPONSE TIME

The Four Core Value Pillars

- **Institutional Knowledge Democratization:** Doctor-verified courses, treatment guides, and feed formulation matrices delivered in vernacular and accessible formats.
- **Operational Digital Transformation:** Replacing loss-prone manual record-keeping with an intuitive, offline-first livestock lifecycle management system (birth, vaccination, breeding, milk/meat yield, sale).
- **Financial Visibility & Unit Economics:** Real-time automated profit/loss calculation per animal batch, milk output tracking, and feed conversion ratio (FCR) analytics.
- **Continuous Veterinary Guardrail:** On-demand access to certified veterinary specialists with historical electronic health records (EHR) attached to every consultation.

2. Core Target Audience & Market Problem Analysis

2.1 Target User Personas

Persona	Profile & Scale	Primary Pain Points	Platform Solution
Commercial Dairy & Cattle Owner	10 to 100+ dairy cows / beef cattle. Semi-automated setup, hired labor.	Missed heat/insemination cycles, mastitis outbreaks, volatile feed costs, lack of individual yield tracking.	Lactation curve tracking, AI/breeding alerts, milk log per cow, doctor-approved mastitis protocols.
Commercial Poultry & Broiler Grower	1,000 to 20,000 birds per batch. Rapid cycle, high mortality risk.	High feed conversion ratio (FCR), sudden flock disease spread, unpredictable batch profitability.	Batch FCR calculators, daily mortality logs, vaccination calendar alerts, biosecurity training courses.
Goat / Sheep Ruminant Farmer	20 to 200 head. Intensive or semi-intensive grazing setup.	PPR / Enterotoxemia mortality, kid mortality, inbreeding, weight gain plateau.	Deworming/vaccination scheduler, weight gain tracking, genetic lineage logging, nutritional eBooks.
Aspiring / New Agropreneur	Urban investors or first-time farm entrants setting up modern farms.	Lack of reliable domain expertise, unverified consultants, poor capital allocation planning.	Foundational video courses, downloadable project ROI Excel models, one-on-one expert onboarding consults.

2.2 Critical Market Problems Solved

- Information Asymmetry & Quackery:** Over 70% of rural livestock keepers rely on informal drug sellers or unlicensed local practitioners, resulting in antibiotic misuse, antibiotic resistance, and high mortality. VETRALINK PRO provides direct access to qualified veterinarians.
- Zero Individual Performance Tracking:** Most farmers treat their herd as an undifferentiated group. They feed non-productive cows the same expensive ration as high-yielding cows. Our ERP isolates bottom-performing animals.
- Fragmented Purchasing Channels:** Farmers currently buy training from random portals, download Excel sheets from unauthorized sources, and contact vets via messaging apps without medical records. VETRALINK PRO consolidates these under one authenticated account.

3. MVP Feature Breakdown: Phase 1 (Launch) vs. Phase 2 (Scale)

Platform Module	Phase 1: Minimum Viable Product (MVP)	Phase 2: Growth & Deep Ecosystem
1. Digital Courses (LMS)	- Video player with secure tokenized streaming (HLS/DASH) - Course catalog (Cattle, Poultry, Goat, Dairy, Nutrition) - Modular chapters, lesson progress tracking - PDF resource attachments per lesson - Direct gateway checkout & instant access	- Automated multilingual subtitles & audio tracks - Quizzes, assessments & verifiable PDF certificates - Live Q&A webinar integration (Zoom/WebRTC) - Interactive community discussion board per lesson

Platform Module	Phase 1: Minimum Viable Product (MVP)	Phase 2: Growth & Deep Ecosystem
2. Digital Store (eBooks & Excel Tools)	- Instant one-click buy & digital fulfillment - In-browser encrypted PDF reader (read-only view) - Watermarked PDF downloads (User Email & Phone stamped) - Pre-built Excel calculators (Feed, FCR, P&L, Loan ROI)	- Interactive in-app Web Calculators (no Excel required) - Dynamic custom guide builder based on herd profile - Bundle offers (eBook + Course + Excel toolkit)
3. Farm Management Software (SaaS ERP)	Animal Master: ID Tag, Breed, Age, Gender, Origin Vaccination & Deworming: Schedule & log records Milk Production: Morning/Evening yields per animal Financial Ledger: Daily feed, medicine expenses & sales Batch Mortality: Bird/flock count tracking Reporting: Monthly P&L and yield summary	Breeding & AI Lifecycle: Heat detection, AI date, pregnancy check, expected calving alerts Feed Inventory Management: Stock depletion alerts Weight Gain & Growth Curve: Automated ADG charting Multi-Farm & Employee Permissions: Staff role management Barcode / QR Code Tag Scanning via mobile camera
4. Subscriptions & Billing	- Basic, Professional & Premium tier enforcement - Monthly & Annual recurring billing via Stripe / bKash / SSLCommerz - Animal record quota limits (e.g. Basic: 25, Pro: 150, Premium: Unlimited) - Automated invoice generation (PDF)	- Custom Enterprise/Cooperative group licensing - Add-on booster packs (Extra 50 animal slots, Extra Vet Credits) - Automated failed-payment dunning flows
5. Tele-Veterinary Services	- Consultation booking form (Species, Symptoms, Photo/Video upload) - Slot booking with Doctor schedule calendar - Doctor Dashboard: Review case, write digital prescription - In-app text chat & prescription PDF generation	- Integrated High-Definition 1-on-1 Video Consultation (WebRTC) - Direct link between Patient Animal Profile and EHR - Emergency SOS broadcast to nearest regional field vets - AI-assisted preliminary triage (symptom matcher)

4. System Architecture & Tech Stack Recommendation

Architectural Philosophy: Modular Monolith to Event-Driven Microservices

To balance rapid time-to-market with high availability and scalability, the MVP is structured as a **Modular Monolith** with distinct domain boundaries (Auth, LMS, Digital Shop, Farm ERP, Billing, Tele-Vet). This prevents microservice orchestration overhead while allowing easy extraction into microservices during Phase 2.

4.1 Recommended Technology Stack

Layer	Technology Selection	Architectural Rationale
Frontend (Web App)	Next.js 14 (React / TypeScript / TailwindCSS)	Server-Side Rendering (SSR) for Course & eBook SEO; High-performance client-side

Layer	Technology Selection	Architectural Rationale
		dashboard for the Farm ERP; responsive design for mobile browsers.
Mobile Applications	React Native (Expo) / Flutter	Shared business logic with Web; excellent offline local caching capabilities (WatermelonDB / SQLite) essential for remote farm environments with poor connectivity.
Backend API Core	Node.js (NestJS) OR Python (FastAPI)	NestJS provides enterprise-grade TypeScript modular architecture, robust dependency injection, and native OpenAPI/Swagger specs. FastAPI ideal if AI triage is added early.
Primary Database	PostgreSQL 16 (AWS RDS / Supabase)	ACID compliance for critical financial ledgers and subscription data; native JSONB support for flexible multi-species biological attributes; PostGIS for farm geo-tagging.
Caching & Queue	Redis (Upstash / Redis Cloud) + BullMQ	Sub-millisecond session and catalog caching; async worker queues for PDF watermarking, payment webhook processing, SMS/WhatsApp notification dispatching.
Object Storage & CDN	Cloudflare R2 + AWS CloudFront	Zero-egress fee storage for heavy video lessons, downloadable eBooks, Excel templates, animal photos, and encrypted prescription PDFs.
Video Streaming	Mux Video / Cloudflare Stream	Automated transcoding, adaptive HLS bitrate streaming, dynamic video watermarking to prevent unauthorized screen recording and course piracy.
Payment Gateways	Stripe (Global) + Local Aggregators (SSLCommerz, bKash, Razorpay)	Supports international cards as well as local mobile financial services (MFS) critical for rural farmers.

4.2 High-Level Architecture Topology



5. User Journeys & End-to-End Flow Diagrams

5.1 Flow 1: Digital Course & Digital Asset Purchase Journey

```

[Visitor Enters Platform]
|
+---> Explores Course / eBook / Excel Catalog (Public Preview & Free Sample)
|
+---> Selects Item & Clicks [Enroll / Buy Now]
|
+---> [Authentication Check]:
|   |└─ Unauthenticated --> Fast SMS OTP / Google Login Popup --> Returns to Cart
|   |└─ Authenticated   --> Proceeds to Checkout
|
+---> Selects Payment Method (Card, Mobile Wallet, NetBanking)
|
+---> Gateway Processes Payment --> Webhook Dispatched to Backend
|
+---> [Backend Verification Worker]:
|   |└─ Updates Order Status to COMPLETED
|   |└─ Creates UserCourseAccess or UserAssetAccess Record
|   |└─ If PDF: Dispatches async background job to stamp Dynamic Watermark
|   |└─ Sends confirmation SMS / WhatsApp / Email with Direct Dashboard Link
|
v
[User Dashboard / My Library]:
|└─ Video Course: Instant playback with chapter tracking and PDF attachments
|└─ eBook: Read in secure web reader OR Download Watermarked PDF
|└─ Excel Tool: One-click download of verified .xlsx template + Video Tutorial
  
```

5.2 Flow 2: Farm Management (SaaS ERP) Daily Operation Journey

```

[Farmer Logs In] ---> Selects Farm Workspace (e.g., "Green Valley Dairy")
|
+---> [Dashboard Overview]: Active Animal Count, Daily Milk Total, Feed Cost, Health Alerts
|
+---> [Action A: Register New Animal]
|   |└─ Inputs Tag #, Species, Breed, DOB, Purchase Price, Photo -> Saved to DB
|
+---> [Action B: Log Daily Milk Production]
|   |└─ Quick-entry grid (Morning / Evening liters per animal or bulk batch)
|   |└─ Instant recalculation of Daily Farm Revenue & Yield Graph
|
+---> [Action C: Health & Vaccination Event]
|   |└─ Selects Tag ID -> Logs Vaccine (e.g. FMD) -> System schedules Next Booster Alert
|
v
[Automated Analytics Engine]:
|└─ Aggregates Feed Incurred vs Milk Produced -> Generates Weekly Profit & Loss Statement
  
```

5.3 Flow 3: Tele-Veterinary Consultation & Prescription Journey

```
[Farmer Notices Sick Animal]
|
+---> Opens [Vet Support] Module in Platform
|
+---> Step 1: Selects Affected Animal from registered ERP inventory (Auto-attaches age/history)
|
+---> Step 2: Fills Chief Complaint, Uploads Photos/Video clips of symptoms & vitals (Temp,
Appetite)
|
+---> Step 3: Selects Consultation Mode (Chat Ticket OR Scheduled Video Call) -> Pays Fee
|
+---> [Doctor Portal]:
|   |--- Doctor gets alert, reviews full Animal Health Profile & uploaded media
|   |--- Conducts Live Call or Chat Consultation
|   |--- Drafts Digital Prescription (Diagnosis, Rx Medicines, Dosage, Precautions)
|   |--- Clicks [Sign & Issue Prescription]
|
v
[Farmer App / SMS]:
|--- Instant notification with link to view / download Official Doctor-Signed PDF Prescription
|--- Prescription treatments automatically appended to that Animal's permanent ERP Medical Record!
```

6. Database Architecture & Entity-Relationship (ER) Overview

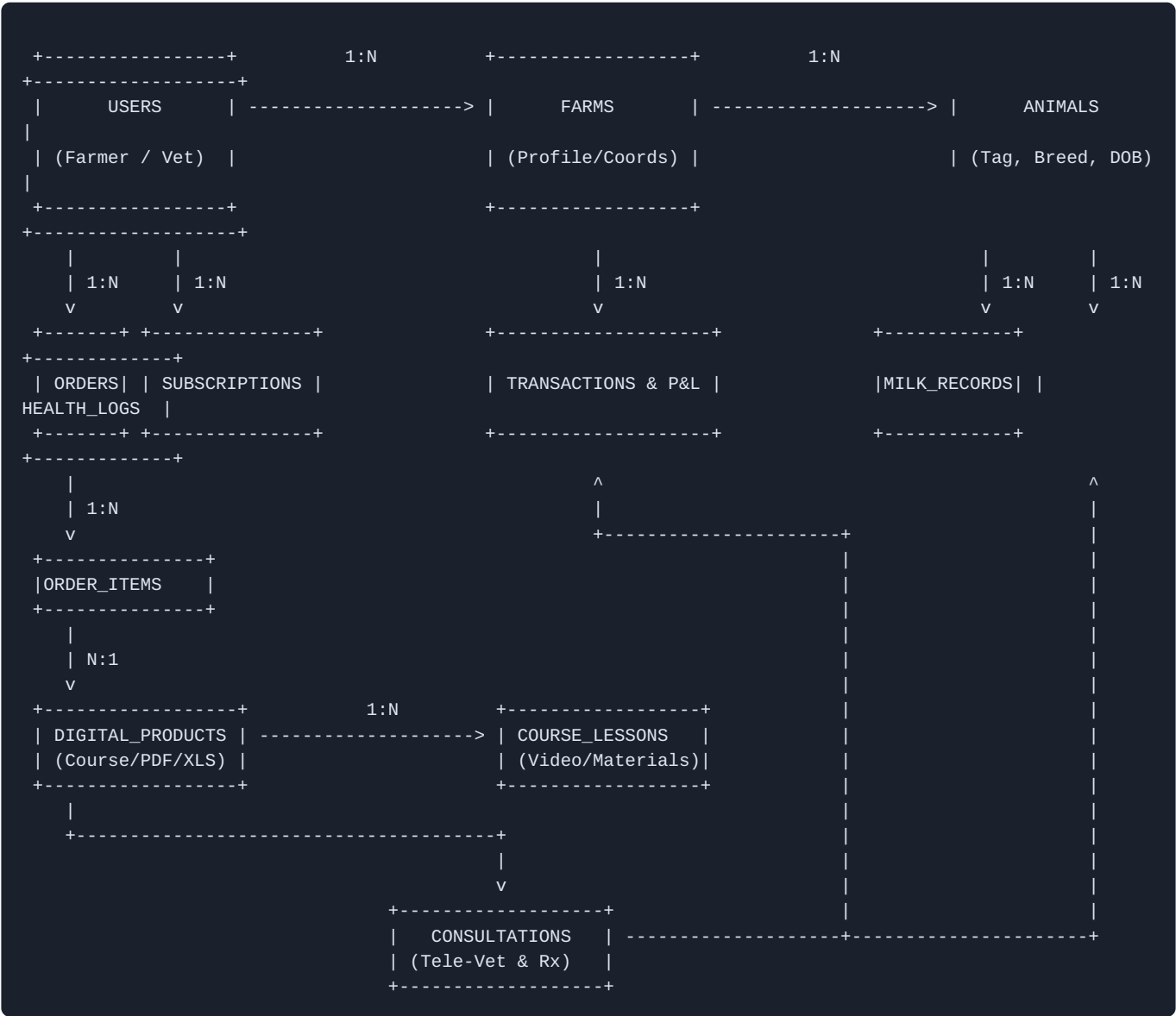
The relational model is designed using third-normal form (3NF) principles with JSONB extensions for species-specific dynamic fields, ensuring zero data duplication, strict referential integrity, and ultra-fast index scans.

6.1 Core Database Entities & Field Specifications

Table Name	Key Attributes & Types	Relationships & Foreign Keys
users	id (UUID PK), phone (VARCHAR Unique), email (VARCHAR), full_name, role (ENUM: farmer, vet, admin), status, created_at	Parent to Farms, Orders, Consultations, Subscriptions.
farms	id (UUID PK), user_id (FK), farm_name, farm_type (dairy, poultry, goat, mixed), location_address, geo_lat, geo_lng	Belongs to users; Parent to animals, farm_logs, transactions.
animals	id (UUID PK), farm_id (FK), tag_number (VARCHAR), species (ENUM), breed, gender, dob, weight_kg, status (active, sold,	Belongs to farms; Parent to vaccinations, treatments, milk_records, breeding_logs.

Table Name	Key Attributes & Types	Relationships & Foreign Keys
	deceased), meta_data (JSONB)	
digital_products	id (UUID PK), title, slug, type (ENUM: course, ebook, excel_tool), price_cents, discount_price_cents, file_url, is_published	Parent to course_lessons, order_items, user_library.
subscriptions	id (UUID PK), user_id (FK), plan_id (FK), status (active, past_due, cancelled), current_period_start, current_period_end, animal_limit	Belongs to users, subscription_plans. Enforces tier limits.
health_records	id (UUID PK), animal_id (FK), event_type (vaccine, deworm, illness, surgery), diagnosis, medicine_name, dosage, administered_by, next_due_date	Belongs to animals. Linked optionally to consultations.
milk_records	id (UUID PK), farm_id (FK), animal_id (FK nullable for flock), record_date, morning_liters, evening_liters, fat_percentage, total_liters	Belongs to farms and animals. Composite index on (farm_id, record_date).
consultations	id (UUID PK), farmer_id (FK), vet_id (FK), animal_id (FK), chief_complaint, media_urls (JSONB[]), status, diagnosis, prescription_pdf_url	Connects users (farmer), users (vet), and animals.

6.2 Entity Relationship Map (ASCII Model)



7. Developer Execution Roadmap & Milestone Sprints

A structured 14-week execution plan engineered for a rapid, robust deployment with zero architectural technical debt.

Milestone	Timeline	Key Deliverables & Developer Scope	Exit / Acceptance Criteria
Sprint 0: Architecture & Setup	Weeks 1 - 2	- Monorepo setup (Turborepo with Next.js, NestJS, Shared Types) - Database schema migration (Prisma/Kysely with PostgreSQL) - CI/CD automated pipeline (GitHub Actions -> Docker -> AWS/Railway) - Authentication service (JWT, SMS OTP, Passwordless)	100% automated tests passing; database seeds ready; auth tokens issued securely.
Sprint 1: Digital	Weeks 3 - 5	- Course catalog & Video streaming module (Mux/Cloudflare) - Digital Store for eBooks & Excel Files	Successful end-to-end checkout; instant video unlock;

Milestone	Timeline	Key Deliverables & Developer Scope	Exit / Acceptance Criteria
Commerce & LMS		- Dynamic PDF watermarking micro-worker (Gotenberg / Python) - Payment Gateway Integration (Stripe + Local MFS Webhooks) - User Library & Order History Dashboard	watermarked PDF download verified.
Sprint 2: Farm ERP Core Engine	Weeks 6 - 9	- Farm Workspace setup & Multi-Species Animal Registry - Daily Milk Production logging & aggregate chart views - Vaccination & Deworming schedule tracker with SMS reminders - Farm Financial Ledger (Feed expense, livestock sales, Net P&L) - Subscription Tier Enforcer (locks features based on active plan)	Farmer can manage 100+ animals, log daily milk/expenses, and generate PDF monthly P&L.
Sprint 3: Tele-Vet Module	Weeks 10 - 12	- Consultation Request pipeline with media attachments - Doctor Admin Dashboard (Case review, patient historical health view) - Digital Prescription Builder (auto-generates signed PDF) - In-app real-time messaging / Socket.io channel	Complete cycle: Farmer books consult -> Doctor writes Rx -> Farmer downloads Rx PDF & ERP logs treatment.
Sprint 4: QA, Security & Launch	Weeks 13 - 14	- End-to-end integration testing & load testing (k6) - Security audit (OWASP Top 10, SQL injection, secure S3 URLs) - Offline PWA caching tuning for low-bandwidth rural access - Production launch & monitoring setup (Sentry, Prometheus)	Production deployment live; payment gateways operational; doctor onboarding complete.

8. Subscription Tiers & Commercial Strategy

Plan	Target Farmer	Included Capabilities	Pricing Model
Starter / Free	Smallholder / Hobbyist (1 - 5 animals)	Basic animal records, community guides, access to buy courses/ eBooks.	Free forever (Lead generator for courses & vet consultations).
Pro Farmer	Commercial Small Herd (Up to 30 animals)	Full Health & Breeding tracking, Milk logging, P&L Reports, SMS vaccination alerts.	\$9 / month (or \$89 / year) recurring subscription.
Commercial Enterprise	Large Dairy / Poultry Farms (Unlimited animals)	Multi-user staff access, advanced batch analytics, priority Tele-Vet response, custom data export.	\$29 / month (or \$289 / year) recurring subscription.

Plan	Target Farmer	Included Capabilities	Pricing Model
Pay-Per-Consult (Add-on)	Any registered farmer needing veterinary help	1-on-1 Certified Vet consultation + Official Digital Prescription.	\$5 - \$15 per consultation (Split: 80% Doctor / 20% Platform fee).

Architectural Sign-off & Recommendation

This production specification delivers the exact balance between immediate commercial cash flow (via digital courses, PDFs, and Excel tools) and recurring high-retention SaaS revenue (via Farm ERP subscriptions and Tele-Vet consultations). By standardizing animal health records directly into veterinary consultations, VETRALINK PRO establishes an unassailable data moat and industry leadership.